

Zhongyu Fan

Sc.M. Student in Biostatistics
Department of Biostatistics, Brown University
121 S Main St, Providence, RI 02903, USA

zhongyu_fan@brown.edu
<https://fan-zhongyu.github.io/>
Phone: 503.515.1758

EDUCATION

Expected 05/2027	Sc.M. in Biostatistics Brown University, School of Public Health Thesis Advisor: Dr. Ying Ma
09/2021–07/2025	B.S. (Honors) in Mathematics with Applied Mathematics University of Nottingham Ningbo China (UNNC), School of Mathematical Sciences Dissertation Title: Bifurcation and Complex Dynamics in a Discrete Predator–Prey Model with a Prey Allee Effect Dissertation Advisors: Dr. Elio Eduardo Espejo Arenas and Dr. Mainul Haque

CONFERENCES

08/2026	Joint Statistical Meetings (JSM), Boston, MA. <i>Student Participant</i>
06/2026	Displacement and Mental Health Symposium, New Haven, CT. <i>Oral presentation, Mental health, suicide risk and behavioral profiles among Ukrainian adolescents: Findings from Wave 1 of the Ukraine Longitudinal Study.</i>

SELECTED RESEARCH EXPERIENCE

Global Mental Health and Flourishing Lab, Brown University **Providence, RI, USA**

11/2025–Present **Research Assistant**

- Investigate associations between war exposure, family-level factors, and mental health outcomes in a longitudinal cohort of approximately 2,200 Ukrainian adolescents and 1,900 parents/caregivers.
- Develop longitudinal and dyadic statistical models using GEE to evaluate associations between war-related exposures and adolescent mental health outcomes, including depression and anxiety.
- Examine intergenerational associations linking parental war exposure and mental health with adolescent mental health outcomes.

The Ma Lab, Brown University

Providence, RI, USA

10/2025–Present **Research Assistant**

- Develop a graph-regularized sparse matrix factorization framework for GWAS summary statistics that incorporates functional genomic annotations.
- Investigate how annotation-informed graph structure can improve latent factor estimation and identification of shared genetic architecture across complex traits.
- Implement the method in R/Rcpp and design simulation and biobank analyses to evaluate statistical performance, robustness, and biological interpretability.

HONORS, AWARDS & CERTIFICATIONS

2025–2026	Todd '87 and Christine Fisher P'17 Graduate Fellowship , Brown University
2025	Heartsaver First Aid CPR AED Certification , American Heart Association
2024	Global Engagement Office Summer Exchange Tuition Waiver Scholarship , UNNC

ACADEMIC SERVICES

06/2024–07/2024	Teaching Assistant , MATH3048: Mathematical Medicine and Biology, UNNC
-----------------	---

LEADERSHIP & PUBLIC SERVICE

09/2023	The 19th Asian Games, Volunteer	Hangzhou, China
04/2022–04/2023	UNNC Campus Ambassador, Vice President	Ningbo, China
06/2022	UNNC Summer Social Practice, Volunteer Teaching Team Leader	Ningbo, China

PROFESSIONAL EXPERIENCE

IQVIA plc **Shanghai, China**

11/2024–01/2025 **Analyst Intern, Management Consulting Department**

HALMA plc

Shanghai, China

07/2024–10/2024 **Business Intelligence and M&A Research Intern, Healthcare Sector**

Beijing AIMEDICINE Co., Ltd. × NYU Shanghai

Shanghai, China

07/2023–08/2023 **Data Analyst, Yangtze River Delta Digital Elderly Care Innovation Research Project**

SKILLS

Computing & Programming: R, Python, C++, Bash/Zsh, Linux, Git, $\LaTeX$, Markdown

Statistical & Machine Learning Tools: PyTorch, Rcpp, tidyverse, ggplot2, lme4, geepack, glmnet, brms

Languages: Mandarin Chinese (Native), English (Proficient).